

Lab Evaluation:

Course Name: Object Oriented Programming				Course Code: CMPE221L		
Assignment Type: Lab				Dated: 23 February 2026		
Semester: 2 nd				Session: 2025		
Lab/Project/Assignment #: 2				CLOs to be covered: CLO1		
Lab Title: Core concepts of Java				Teacher Name: Mr. Mujtaba Kamal		
CLO 1	The manual provides a comprehensive guide to the core concepts of Java, based on the standard curriculum. Java is a popular, object-oriented programming language used for developing mobile apps, web apps, desktop apps, games, and much more.					
Levels (Marks)	Level 1	Level 2	Level 3	Level 4	Level 5	Level 6
(10)						
Total						/10

Rubrics for Current Lab Evaluation

Scale	Marks	Level	Rubric
Excellent	9-10	L1	Submitted all lab tasks, BONUS task, have good understanding.
Very Good	7-8	L2	Submitted the lab tasks but have good understanding
Good	5-6	L3	Submitted the lab tasks but have weak understanding.
Basic	3-4	L4	Submitted the lab tasks but have no understanding.
Barely Acceptable	1-2	L5	Submitted only one lab task.
Not Acceptable	0	L6	Did not attempt

LAB No.2

Java Programming Manual: Fundamentals

1. Syntax

Java code is written inside **classes**. The filename must match the class name followed by .java.

```
public class Main {  
    public static void main(String[] args) {  
        System.out.println("Hello World");  
    }  
}
```

- **public class Main:** Every line of code that runs in Java must be inside a class.
- **main() method:** This is the entry point of any Java program. Any code inside its curly brackets {} will be executed.

2. Output

To print text to the console, use the System.out.println() or System.out.print() methods.

- **println():** Adds a new line after the output.
- **print():** Does not insert a new line at the end.

Java

```
System.out.println("Line One");  
System.out.print("This stays on ");  
System.out.print("the same line.");
```

3. Comments

Comments are used to explain code and make it more readable. They are ignored by the compiler.

- **Single-line:** Starts with //.
- **Multi-line:** Starts with /* and ends with */.

Java

```
// This is a single-line comment  
/* This is a multi-line comment  
   used for longer descriptions */
```

4. Variables

Variables are containers for storing data values. In Java, you must specify the type.

- **String:** Stores text (e.g., "Mujtaba").
- **int:** Stores integers (e.g., 123).
- **float:** Stores floating-point numbers (e.g., 19.99).
- **char:** Stores single characters (e.g., 'a').
- **boolean:** Stores true or false.

Syntax: type variableName = value;

Java

```
String name = "Mujtaba";
```

```
int myNum = 15;
```

5. Data Types

Data types are divided into two groups:

1. **Primitive data types:** Includes byte, short, int, long, float, double, boolean, and char.
2. **Non-primitive data types:** Includes String, Arrays, and Classes.

6. Type Casting

Type casting is assigning a value of one primitive data type to another type.

- **Widening Casting (automatically):** Smaller type to a larger type size.

byte -> short -> char -> int -> long -> float -> double

```
int myInt = 9;
```

```
double myDouble = myInt; // Automatically casting: int to double
```

- **Narrowing Casting (manually):** Larger type to a smaller size type.

double -> float -> long -> int -> char -> short -> byte

Java

```
double myDouble = 9.78d;
```

```
int myInt = (int) myDouble; // Manual casting: double to int
```

7. Operators

Operators are used to perform operations on variables and values.

- **Arithmetic:** +, -, *, /, %, ++, --
- **Assignment:** =, +=, -=, *=, /=
- **Comparison:** ==, !=, >, <, >=, <=
- **Logical:** && (and), || (or), ! (not)

8. Strings

A String in Java is an object that contains a collection of characters.

- **Length:** `txt.length()`
- **Case:** `txt.toUpperCase()`, `txt.toLowerCase()`
- **Find:** `txt.indexOf("word")`
- **Concatenation:** Using the `+` operator or `concat()`.

```
String firstName = "John ";
```

```
String lastName = "Doe";
```

```
System.out.println(firstName.concat(lastName));
```

9. Math Class:

The Math class allows you to perform mathematical tasks on numbers.

- `Math.max(x, y)`: Finds the highest value.
- `Math.min(x, y)`: Finds the lowest value.
- `Math.sqrt(x)`: Returns the square root.
- `Math.abs(x)`: Returns the absolute (positive) value.
- `Math.random()`: Returns a random number between 0.0 and 1.0.

10. Booleans

A boolean type is declared with the `boolean` keyword and can only take the values `true` or `false`. This is mostly used for conditional testing.

```
Java
```

```
boolean isJavaFun = true;
```

```
System.out.println(10 > 9); // Returns true
```

11. If...Else

Use conditional statements to perform different actions based on different conditions.

- **if:** Executes code if a condition is true.
- **else:** Executes code if the condition is false.
- **else if:** Specifies a new condition if the first is false.

```
int time = 20;
```

```
if (time < 18) {
```

```
    System.out.println("Good day.");
```

```
} else {
```

```
    System.out.println("Good evening.");
```

```
}
```

12. Java Switch

Use the switch statement to select one of many code blocks to be executed.

Java

```
int day = 2;

switch (day) {

    case 1:

        System.out.println("Monday");

        break;

    case 2:

        System.out.println("Tuesday");

        break;

    default:

        System.out.println("Looking forward to the Weekend");

}
```

- The break keyword breaks out of the switch block.
- The default keyword specifies code to run if there is no case match.

This lab series is designed to take you from basic syntax to intermediate logical problem-solving. Each task focuses on a specific set of concepts from your manual.

Task 1: The Geometry Calculator (Operators & Math)

Objective: Use Math methods and arithmetic operators.

- Create a program that stores the radius of a circle as a double.
- Calculate the area ($\text{Area} = \pi \times r^2$). Use `Math.PI` and `Math.pow()`.
- Print the result. Then, use **Type Casting** to print the area as a whole integer.

Task 2: String Analysis Tool (Strings)

Objective: Manipulate and analyze text data.

- Create a String variable `fullName` containing your first and last name.
- Print the length of the string.
- Convert the name to all uppercase.
- Use `indexOf()` to find the position of the space between the names.

Task 3: Automatic Grade Converter (Type Casting)

Objective: Practice Narrowing Casting and Output.

- Declare a double variable `percentage = 85.99`.
- Manually cast this percentage into an int.
- Print: "Your raw score is: 85.99% but your rounded grade is: 85%".

Task 4: The Logical Gatekeeper (Logical Operators)

Objective: Use `&&`, `||`, and `!`.

- Create two boolean variables: `hasID` and `isOver18`.
- Write an if statement that prints "Access Granted" only if both are true.
- Write another statement that prints "Special Guest" if they have an ID **OR** they are over 18.

Task 5: Random Number Matcher (Math & Booleans)

Objective: Use `Math.random()` and comparisons.

- Generate a random number between 1 and 10. *Hint: Use `(int)(Math.random() * 11)`.*
- Declare a variable `userGuess = 5`.
- Print true if the random number matches the guess, otherwise print false.

Task 6: Personalized Greeting (Strings & Concatenation)

Objective: Use different methods of string joining.

- Create variables for greeting, firstName, and lastName.
- Use the + operator to join them for one output.
- Use the .concat() method to join them for a second output.

Task 7: Time-of-Day Identifier (If...Else If)

Objective: Use conditional chains.

- Store an int variable hour (0 to 23).
- If hour is between 5 and 11, print "Good Morning".
- Else if hour is between 12 and 17, print "Good Afternoon".
- Else if hour is between 18 and 23, print "Good Evening".
- Use else to handle any other numbers as "Invalid Hour".

Task 8: The Currency Switcher (Switch Case)

Objective: Practice multi-way branching.

- Create a variable String currency = "USD".
- Use a switch statement to print the full name of the currency:
 - "USD" -> "United States Dollar"
 - "GBP" -> "British Pound"
 - "EUR" -> "Euro"
 - **Default** -> "Unknown Currency"

Task 9: Comparison & Increment Lab (Operators)

Objective: Understand how values change.

- Declare int x = 10 and int y = 20.
- Use a comparison operator to print if x is not equal to y.
- Apply the increment operator ++ to x.
- Print the new value of x and check if it is now greater than 10.